

MODULE-3

STATISTICAL INFERENCE-1

Population:

A large collection of individuals or attributes or numerical data can be regarded as a population or universe.

Sample:

A finite subset of the universe is called a sample.

The number of individuals in a sample is called a sample size. Sample size is denoted by 'n'.

Small and large sample:

If $n \leq 30$, then sample is said to be small.

If $n > 30$, then sample is said to be large sample.

Sampling:

The process of selecting a sample from the population is called as sampling.

Random Sampling:

The selection of an individual or item from the population in such a way that each has the same chance of being selected is called as random sampling.

Suppose we take a sample of size n from a finite population of size N , then we will have ${}^N C_n$ possible samples.

Random sampling is a technique in which each of the ${}^N C_n$ samples has an equal chance of being selected.

Sampling with replacement:

Sampling where a member of the population may be selected more than once is called as sampling with replacement.

Sampling without replacement:

Sampling where a member of the population cannot be chosen more than once is called as sampling without replacement.

Simple sampling:

Simple sampling is a special case of random sampling in which trials are independent and the probability of success is a constant.

E.g: selecting 25 employees out of 250 employees in a company.

Statistic:

The statistical constants of the population such as mean (μ), standard deviation (σ) etc., are called the parameters. These statistical constants for the sample drawn from the given population i.e., mean ($\bar{x}$), standard deviation (s) etc., are called the statistic.

Statistical inference:

Sampling aims at gathering the maximum information about the population with the minimum effort, cost and time.

The logic of the sampling theory is the logic of induction in which we pass from a particular (sample) to general (population). Such a generalisation from sample to population is called statistical inference.

Sampling distribution:

Consider all possible samples of size n which can be drawn from a given population at random. For each sample, we can compute the mean. The means of the samples will not be identical. If we group these different means according to their frequencies, the frequency distribution so formed is known as sampling distribution of the mean. Similarly we can have sampling distribution of the standard deviation etc.

Standard error:

The standard deviation of the sampling distribution is called the standard error (S.E)

The reciprocal of the standard error is called precision.

Statistical hypothesis:

To reach decisions about populations on the basis of sample information, we make certain assumptions about the populations involved. Such assumptions, which may or may not be true are called statistical hypothesis.

or

A statistical hypothesis is an assertion or conjecture (assumption or guess) about the distribution of one or more random variables.

Null hypothesis:

The hypothesis formulated for the purpose of its rejection under the assumption that it is true is called Null hypothesis denoted by H_0 .

The complementary to the null hypothesis is called alternative hypothesis and is denoted by H_1 .

Errors in Sampling:

While testing a hypothesis we usually may commit two types of errors. They are classified as Type-I and Type-II errors.

Type-I error:

Rejecting the hypothesis when it should be accepted is called Type-I error and is denoted by α .

The probability of correct decision in this case is $1 - \alpha$.

Type-II error:

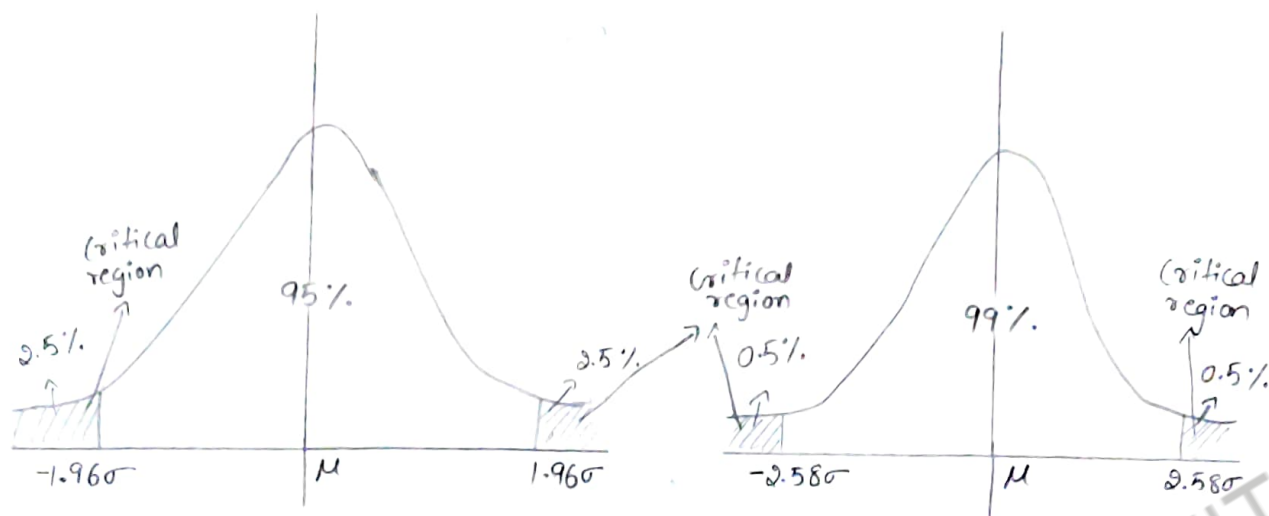
Accepting the hypothesis when it should be rejected is called Type-II error and is denoted by β .

The probability of correct decision in this case is $1 - \beta$.

	Accepting the hypothesis	Rejecting the hypothesis
Hypothesis true	Correct decision	Wrong decision (Type I error)
Hypothesis false	Wrong decision (Type II error)	Correct decision

In order to minimize both these types of errors we need to increase the sample size.

Level of Significance or Significance Level:



The probability level, below which leads to the rejection of the hypothesis is known as the significance level or level of significance.

The region in which a sample value falling is rejected is known as the critical region.

We generally take two critical regions which cover 5% and 1% areas of the normal curve.

The shaded portion in the figure corresponds to 5% level of significance and 1% level of significance.

NOTE: * The sampling distribution of large samples is assumed to be a normal distribution.

* 5% or 1% are called significance level

Test of Significance:

The process which helps us to decide about the acceptance or rejection of the hypothesis is called the test of significance.

Confidence interval, confidence limits and confidence coefficients:

Suppose that the sampling distribution of a statistic S is normal with mean μ and standard deviation σ , then the sample statistic S can be expected to lie in the interval $(\mu - 1.96\sigma, \mu + 1.96\sigma)$ for 95% times and $(\mu - 2.58\sigma, \mu + 2.58\sigma)$ for 99% times. Because of this we call 95% confidence interval and 99% confidence interval respectively for estimation of μ .

The ends of the confidence interval (i.e., $S \pm 1.96\sigma$) are called 95% confidence limits for S . Similarly the ends of the confidence interval (i.e., $S \pm 2.58\sigma$) are called 99% confidence limits for S .

The numbers 1.96, 2.58 etc., are called confidence coefficients.

Confidence coefficients are denoted by Z_c .

Confidence interval:

From the table of normal we find that 95% of area lies between $Z = -1.96$ and $Z = 1.96$. This is confidence interval in 5% level of significance and is denoted by $Z_{0.05}$.

Similarly 99% of area lies between -2.58 & 2.58 . This is confidence interval in 1% level of significance and is denoted by $Z_{0.01}$.

Probable limits formula:

$$\text{Probable limits} = p \pm Z_c \sqrt{\frac{pq}{n}}$$

where $p \rightarrow$ probability of success

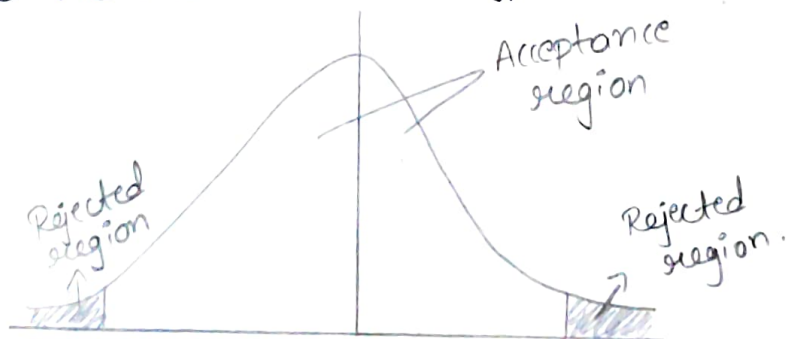
$q \rightarrow$ probability of failure

$Z_c \rightarrow$ critical value.

$n \rightarrow$ sample size

Two-Tailed Test: A test of a hypothesis, where the area of rejection is on both sides of the sampling distribution is called a two-tailed test.

It is a non-directional hypothesis.



One-tailed test:

A test of a statistical hypothesis, where the area of rejection is on only one side of the sampling distribution is called one-tailed test.

It is a directional hypothesis.



Illustrative examples on null hypothesis and alternate hypothesis:

For example, if we want to test the null hypothesis ($H_0: \mu = \mu_0$) then the H_1 could be

$H_1: \mu \neq \mu_0$	Two tailed alternatives
$H_1: \mu > \mu_0$	Right-tailed alternatives
$H_1: \mu < \mu_0$	Left-tailed alternatives

Example 1:

A teacher claims that the median time to do a particular type of statistics problems is at most 3 minutes, but her students believe that the median time is more than 3 minutes. A random sample of 10 students completed the problem in the following times (in minutes) i.e., 2.5, 2, 4, 4.5, 4, 2.5, 4.5, 3, 3.5, 5

Use the sign test with 5% level of significance to test the teacher's claim.

Soln: Null Hypothesis: $H_0: \eta \leq 3$

Alternate hypothesis: $H_1: \eta > 3$.

Example 2:

A milling process currently produces on average of 3% defectives. You are interested in showing that a simple adjustment on a machine will decrease the proportion of defectives produced in the milling process.

Soln: Null hypothesis: $H_0: p \geq 0.03$ or $p = 0.03$

Alternate hypothesis: $H_1: p < 0.03$.

NOTE:

- 1) 1% level of significance = 99% confidence level
- 2) 5% level of significance = 95% confidence level.
- 3) Standard error of the proportion of success

$$= \sqrt{n \cdot \frac{p}{n} \cdot \frac{q}{n}} = \sqrt{\frac{pq}{n}}$$

Problems:

(MQP-2)

- 1) A coin was tossed 1000 times and head turned up 540 times. Decide on the hypothesis that the coin is unbiased.

or

A coin was tossed 1000 times and head turned up 540 times. Can we infer that coin is unbiased at 1% level of significance.

Soln: Let us suppose that the coin is unbiased.

p = probability of getting a head in one toss = $\frac{1}{2}$

Since $p+q=1$, $q=\frac{1}{2}$

Expected no of heads, $np = 1000 \times \frac{1}{2} = 500$

Actual no of heads, $x = 540$

The difference, $x - np = 540 - 500 = 40$

$$\text{Consider, } Z = \frac{x - np}{\sqrt{npq}} = \frac{40}{\sqrt{1000 \times \frac{1}{2} \times \frac{1}{2}}} = \frac{40}{\sqrt{250}}$$

$$Z = \frac{40}{15.8114} = 2.53 < 2.58$$

As $z < 2.58$, the hypothesis is accepted at 1% level of significance.

Thus we can say that the coin is unbiased.

[* unbiased coin has equal probability of heads or tails]

(MQP-1)

2. A die was thrown 9000 times and a throw of 3 or 4 was observed 3240 times, show that the die cannot be regarded as an unbiased one.

Soln: Probability of getting 3 or 4 in a single throw is, $p = \frac{2}{6} = \frac{1}{3}$

$$q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

Expected no of successes, $np = 9000 \times \frac{1}{3} = 3000$

observed no of successes, $x = 3240$

The difference, $x - np = 3240 - 3000 = 240$.

$$\text{consider, } Z = \frac{x - np}{\sqrt{npq}} = \frac{240}{\sqrt{9000 \times \frac{1}{3} \times \frac{2}{3}}} = \frac{240}{\sqrt{2000}} = \frac{240}{44.7214}$$

$$Z = 5.37 > 2.58$$

As $Z > 2.58$, the hypothesis is rejected at 1% level of significance.

we conclude that the die is biased.

3) Result extracts revealed that in a certain school over a period of five years 725 students had passed and 615 students had failed. Test the hypothesis that success & failure are in equal proportions.

Soln: Total no of students = $725 + 615 \Rightarrow \boxed{1340 = n}$

Suppose that success & failure are in equal proportion. Then $p = \frac{1}{2}$ and $q = \frac{1}{2}$.

Expected no of successes, $np = 1340 \times \frac{1}{2} = 670$

Observed no of successes, $x = 725$.

The difference, $x - np = 725 - 670 = 55$

$$\sigma = \sqrt{npq} = \sqrt{1340 \times \frac{1}{2} \times \frac{1}{2}} = \sqrt{335} = 18.3030.$$

$$\text{Consider, } Z = \frac{x - np}{\sqrt{npq}} = \frac{55}{18.3030} = 3.005 > 2.58.$$

As $Z > 2.58$ the hypothesis is rejected at 1% level of significance.

Hence we conclude that success & failure are not in equal proportions.

4. In a hospital 230 females & 270 males were born in a year. Do these figures confirm the hypothesis that sexes are born in equal proportions?

Soln: Given, Total = $230 + 270 = 500$.

Let us suppose that sexes are born in equal proportions then $p = \frac{1}{2}$

$$\therefore q = \frac{1}{2}$$

Expected no of successes, $np = 500 \times \frac{1}{2} = 250$

Actual no of successes, $x = 270$.

The difference, $x - np = 270 - 250 = 20$

$$\sigma = \sqrt{npq} = \sqrt{500 \times \frac{1}{2} \times \frac{1}{2}} = \sqrt{125} = 11.1803$$

$$\text{Consider, } Z = \frac{x - np}{\sqrt{npq}} = \frac{20}{11.1803} = 1.79 < 1.96$$

As $Z < 1.96$, the hypothesis is accepted at 5% level of significance.

Thus, we conclude that sexes are born in equal proportion.

5) (M.Q.P.-II)

In 324 throws of a six faced die, an odd number turned up 181 times. Is it reasonable to think that the die is unbiased one at 5% level of significance?

Soln: Given, $n = 324$

Probability of getting an odd number in a single throw is, $p = \frac{3}{6} = \frac{1}{2}$.

$$\Rightarrow q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$$

Expected no of successes, $np = 324 \times \frac{1}{2} = 162$

observed no of successes, $x = 181$

The difference, $x - np = 181 - 162 = 19$

$$\sigma = \sqrt{npq} = \sqrt{324 \times \frac{1}{2} \times \frac{1}{2}} = \sqrt{81} = 9.$$

$$\text{Consider, } Z = \frac{x - np}{\sqrt{npq}} = \frac{19}{9} = 2.11 > 1.96$$

As $Z > 1.96$, the hypothesis is rejected at 5% level of significance.

Hence we conclude that the die is biased.

(Q.P-1)

6) A coin was tossed 400 times and head turned up 216 times. Test the hypothesis that the coin is unbiased at 5% level of significance.

Soln: Let us suppose that the coin is unbiased.

Probability of getting head in one toss, $p = \frac{1}{2}$

Since $p+q=1$, $q = \frac{1}{2}$

Expected no of heads, $np = 400 \times \frac{1}{2} = 200$

Actual no of heads, $x = 216$.

The difference, $x - np = 216 - 200 = 16$.

$$\sigma = \sqrt{npq} = \sqrt{400 \times \frac{1}{2} \times \frac{1}{2}} = \sqrt{100} = 10.$$

$$\text{consider, } z = \frac{x - np}{\sqrt{npq}} = \frac{16}{10} = 1.6 < 1.96$$

As $z < 1.96$, the hypothesis is accepted at 5% level of significance.

Thus we conclude that the coin is unbiased.

Problems on probable limits:

- 1) A sample of 900 days was taken in a coastal town and it was found that on 100 days the weather was very hot. obtain the probable limits of the percentage of very hot weather.

Soln: Probability of very hot weather, $p = \frac{100}{900} = \frac{1}{9} = 0.111$

$$\therefore q = \frac{8}{9}$$

$$\text{Probable limits} = p \pm 2.58 \sqrt{pq/n}$$

$$= 0.111 \pm (2.58) \sqrt{\frac{1}{9} \times \frac{8}{9} \times \frac{1}{900}}$$

$$= 0.111 \pm 0.027$$

$$= (0.111 - 0.027) \text{ and } (0.111 + 0.027)$$

$$= 0.084 \text{ and } 0.138$$

$\therefore$ Probable limits of very hot weather is 8.4% and 13.8%.

- 2) In a sample of 500 men it was found that 60% of them had over weight. what can we infer about the proportion of people having over weight in the population?

Soln: Probability of persons having over weight, $p = \frac{60}{100} = 0.6$

$$\therefore q = 1 - p = 0.4$$

$$\text{Probable limits} = p \pm 2.58 \sqrt{\frac{pq}{n}}$$

$$= 0.6 \pm 2.58 \sqrt{\frac{(0.6)(0.4)}{500}}$$

$$= 0.6 \pm 0.0565$$

$$= (0.6 - 0.0565) \text{ and } (0.6 + 0.0565)$$

$$= 0.5435 \text{ and } 0.6565$$

Thus the probable limits of people having over weight is 54.35% to 65.65%.

3) A survey was conducted in a slum locality of 2000 families by selecting a sample of size 800. It was revealed that 180 families were illiterates. Find the probable limits of the illiterate families in the population of 2000.

Soln: Probability of illiterate families, $p = \frac{180}{800} = 0.225$
 $\therefore q = 0.775$

$$\text{Probable limits} = p \pm 2.58 \sqrt{\frac{pq}{n}}$$

$$= 0.225 \pm 2.58 \sqrt{\frac{(0.225)(0.775)}{800}}$$

$$= 0.225 \pm 0.038$$

$$= (0.225 - 0.038) \text{ and } (0.225 + 0.038)$$

$$= 0.187 \text{ and } 0.263$$

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∴ The probable limits of illiterate families in the population of 2000 is.

$$(0.187)2000 \text{ and } (0.263)2000$$

$$\Rightarrow 374 \text{ and } 526.$$

Thus 374 to 526 are probably illiterate families.

4) A sample of 100 days is taken from meteorological records of a certain district and 10 of them are found to be foggy. what are the probable limits of the percentage of foggy days in the district.

Soln: Probability of foggy days, $p = \frac{10}{100} = 0.1$

$$\therefore q = 0.9$$

$$\begin{aligned} \text{Probable limits} &= p \pm 2.58 \sqrt{\frac{pq}{n}} \\ &= 0.1 \pm 2.58 \sqrt{\frac{(0.1)(0.9)}{100}} \\ &= 0.1 \pm 0.0774 \\ &= (0.1 - 0.0774) \text{ and } (0.1 + 0.0774) \\ &= 0.0226 \text{ and } 0.1774 = 2.26\% \text{ \& } 17.74\% \end{aligned}$$

Thus the percentage of foggy days lies between 2.26 and 17.74.

5) A random sample of 500 apples was taken from a large consignment and 65 were found to be bad. Estimate the proportion of bad apples in the consignment as well as the standard error of the estimate. Also find the percentage of bad apples in the consignment.

Soln: Probability of bad apples, $p = \frac{65}{500} = 0.13$

$$\therefore q = 0.87$$

$$S.E = \sqrt{\frac{pq}{n}} = \sqrt{\frac{(0.13)(0.87)}{500}} = 0.015$$

$$\begin{aligned}\text{Probable limits} &= p \pm 2.58 \sqrt{\frac{pq}{n}} \\ &= 0.13 \pm 2.58 (0.015) \\ &= 0.13 \pm 0.0387 \\ &= (0.13 - 0.0387) \text{ and } (0.13 + 0.0387) \\ &= 0.0913 \text{ and } 0.1687 \\ &= 9.13\% \text{ and } 16.87\%.\end{aligned}$$

Thus the required percentage of bad apples in the consignment lies b/w 9.13 and 16.87.

6) In a locality of 18000 families a sample of 840 families was selected at random. of these 840 families, 206 families were found to have monthly income of Rs. 2500 or less. It was desired to estimate how many of the 18,000 families have monthly income of Rs. 2500 or less. within what limits would you place your estimate.

Soln: Probability of families having monthly income of Rs. 2500 or less, $p = \frac{206}{840} = 0.245$

$$\therefore q = 0.755.$$

$$S.E = \sqrt{\frac{pq}{n}} = \sqrt{\frac{(0.245)(0.755)}{840}} = 0.015$$

$$\begin{aligned} \text{Probable limits} &= p \pm 2.58 \sqrt{\frac{pq}{n}} \\ &= 0.245 \pm 2.58 (0.015) \\ &= 0.245 \pm 0.0387 \\ &= (0.245 - 0.0387) \text{ and } (0.245 + 0.0387) \\ &= 0.2063 \text{ and } 0.2837 \\ &= 20.63\% \text{ and } 28.37\%. \end{aligned}$$

Hence the probable limits in respect of 18,000 families is
 $(0.2063) 18000$ and $(0.2837) 18000$

Thus 3713 to 5107 families are likely to have monthly income of Rs. 2500 or less.

NOTE:

LEVEL OF SIGNIFICANCE

Types of Test	1%	5%	10%
Two tailed test	± 2.58	± 1.96	± 1.645
one tailed test	± 2.33	± 1.645	± 1.28

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Test of significance for proportion (or) comparison of large samples:

Test of significance of single proportion:

To test the significant difference b/w the sample proportion p and the population proportion P , we use the statistic,

$$Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \quad \text{where } P + Q = 1$$

$n \rightarrow \text{sample size}$

The formulated Null and alternative hypothesis are

$H_0: P = \text{a specified value}$

$H_1: P \neq \text{a specified value.}$

Test of significance of difference b/w two sample proportions:

To test the significance of the difference b/w the samples proportions, the test statistic under the null hypothesis H_0 that there is no significance difference b/w the 2 sample proportions,

$$Z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{p_1 - p_2}{c}; c = \sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

where $P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ or $P = \frac{x_1 + x_2}{n_1 + n_2}$ and $P + Q = 1$

p_1 and p_2 are the sample proportions in

respect of an attribute corresponding to two large samples of size n_1 and n_2 drawn from the two populations.

Problems:

1) If a sample of 300 units of a manufactured product 65 units were found to be defective and in another sample of 200 units, there were 35 defectives. Is there significant difference in the proportion of defectives in the samples at 5% level of significance?

Soln: Null hypothesis: $H_0: P_1 = P_2$ i.e., there is no significant difference in the proportion of defectives.

Alternative hypothesis: $H_1: P_1 \neq P_2$

Given $n_1 = 300$, $n_2 = 200$; $x_1 = 65$, $x_2 = 35$

In a sample of 300 units, 65 units were found to be defective. $\therefore p_1 = \frac{65}{300} = 0.2167$

In a sample of 200 units, 35 units were found to be defective. $\therefore p_2 = \frac{35}{200} = 0.175$

$$\text{WKT, } P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{65 + 35}{300 + 200} = \frac{100}{500} = \underline{\underline{0.2}}$$

Since $Q = 1 - P = 1 - 0.2$

$$\Rightarrow Q = 0.8$$

$$e = \sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{(0.2)(0.8) \left(\frac{1}{300} + \frac{1}{800} \right)}$$

$$e = \sqrt{0.001333} = 0.0365$$

$$\therefore Z = \frac{p_1 - p_2}{e} = \frac{0.2167 - 0.175}{0.0365} = 1.14 < 1.96$$

As $|z| < 1.96$, the hypothesis is accepted at 5% level of significance.

Hence we conclude that there is no significant difference in the proportion of defectives.

2) In a large city A, 20% of a random sample of 900 school boys had a slight physical defect. In another large city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference b/w the proportion significant?

Soln: Null hypothesis: $H_0: P_1 = P_2$ i.e., there is no significant difference.

Alternative hypothesis: $H_1: P_1 \neq P_2$

Given $n_1 = 900$, $n_2 = 1600$; $p_1 = \frac{20}{100} = 0.2$, $p_2 = \frac{18.5}{100} = 0.185$

$$x_1 = 20\% \text{ of } 900 = \frac{20}{100} \times 900 = 180$$

$$x_2 = 18.5\% \text{ of } 1600 = \frac{18.5}{100} \times 1600 = 296$$

$$\text{WKT, } p = \frac{x_1 + x_2}{n_1 + n_2} = \frac{180 + 296}{900 + 1600} = \frac{476}{2500}$$

$$p = 0.1904$$

$$\text{Since } q = 1 - p = 1 - 0.1904 = 0.8096.$$

$$c = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{(0.1904)(0.8096) \left(\frac{1}{900} + \frac{1}{1600} \right)}$$

$$c = \sqrt{0.00026762} = 0.0163$$

$$\therefore z = \frac{p_1 - p_2}{c} = \frac{0.2 - 0.185}{0.0163} = 0.92 < 1.96$$

As $|z| < 1.96$, the hypothesis is accepted at 5% level of significance.

Hence we conclude that there is no significant difference.

3) Before an increase in excise duty on tea, 800 persons out of a sample of 1000 persons were found to be tea drinkers. After an increase in excise duty, 800 people were tea drinkers in a sample of 1200 people. Test whether there is a significant decrease in the consumption of tea after the increase in excise duty at 5% level of significance.

Soln: Null hypothesis: $H_0: P_1 = P_2$ i.e., there is no significant difference in the consumption of tea before and after the increase in excise duty.

Alternative hypothesis: $H_1: P_1 \neq P_2$. ($P_1 > P_2$ or $P_2 < P_1$)

Given $n_1 = 1000$, $n_2 = 1200$; $x_1 = 800$, $x_2 = 800$

$$\therefore p_1 = \frac{800}{1000} = 0.8, \quad p_2 = \frac{800}{1200} = 0.6667.$$

$$\text{WKT, } P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{800 + 800}{1000 + 1200} = 0.7272.$$

$$\text{Since } Q = 1 - P = 1 - 0.7272 = 0.2727.$$

$$e = \sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{(0.7272)(0.2727) \left(\frac{1}{1000} + \frac{1}{1200} \right)}$$

$$e = \sqrt{0.0003636} = 0.0191$$

$$\therefore Z = \frac{p_1 - p_2}{e} = \frac{0.8 - 0.6667}{0.0191} = 6.98 > 1.96$$

Since $|Z| > 1.96$, the hypothesis is rejected at 5% level of significance.

Hence we conclude that there is a significant decrease in the consumption of tea due to increase in excise duty.

4) In a sample of 600 men from a certain city, 450 are found smokers. In other sample of 900 men from another city, 450 are smokers. Do the data indicate that the cities are significantly different with respect to the habit of smoking among men. Test at 5% significance level.

Soln: Null hypothesis: $H_0: P_1 = P_2$ i.e., there is no significant difference.

Alternative hypothesis: $H_1: P_1 \neq P_2$

Given $n_1 = 600$, $n_2 = 900$; $x_1 = 450$, $x_2 = 450$

$$\therefore p_1 = \frac{450}{600} = 0.75 ; p_2 = \frac{450}{900} = 0.5$$

$$\text{WKT, } P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{450 + 450}{600 + 900} = \frac{900}{1500} = \underline{\underline{0.6}}$$

$$\text{Since } Q = 1 - P = 1 - 0.6 \\ \Rightarrow \underline{\underline{Q = 0.4}}$$

$$c = \sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{(0.6)(0.4) \left(\frac{1}{600} + \frac{1}{900} \right)} = \sqrt{0.000667}$$

$$c = 0.0258$$

$$\therefore Z = \frac{p_1 - p_2}{c} = \frac{0.75 - 0.5}{0.0258} = 9.69 > 1.96$$

As $|z| > 1.96$, the hypothesis is rejected at 5% level of significance.

Hence we conclude that there is a significant difference.

5) One type of air craft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total of 200 flights. Is there a significance difference in the two types of air craft's so far as engine defects are concerned?

Test at 5% significance level.

Soln: Null hypothesis: $H_0: P_1 = P_2$ i.e., there is no significant difference.

Alternative hypothesis: $H_1: P_1 \neq P_2$.

Given $n_1 = 100$, $n_2 = 200$; $x_1 = 5$, $x_2 = 7$.

$$\therefore p_1 = \frac{5}{100} = 0.05 \quad ; \quad p_2 = \frac{7}{200} = 0.035$$

$$\text{WKT, } P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{5 + 7}{100 + 200} = \frac{12}{300} = 0.04$$

$$\text{Since } Q = 1 - P = 1 - 0.04$$

$$\Rightarrow \underline{Q = 0.96.}$$

$$e = \sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{(0.04)(0.96) \left(\frac{1}{100} + \frac{1}{200} \right)}$$

$$e = \sqrt{0.000576} = 0.024$$

$$Z = \frac{p_1 - p_2}{e} = \frac{0.05 - 0.035}{0.024} = 0.625$$

As $|Z| < 1.96$, the hypothesis is accepted at 5% level of significance.

Hence we conclude that there is no significant difference.

6) A machine produced 16 defective articles in a batch of 500. After overhauling it produced 3 defectives in a batch of 100. Has the machine improved.

Soln: Null hypothesis: $H_0: P_1 = P_2$ i.e., the machine has not improved after overhauling.

Alternative hypothesis: $H_1: P_1 \neq P_2$.

Given $n_1 = 500$, $n_2 = 100$; $x_1 = 16$, $x_2 = 3$

$$\therefore p_1 = \frac{16}{500} = 0.032; \quad p_2 = \frac{3}{100} = 0.03.$$

WKT, $p = \frac{x_1 + x_2}{n_1 + n_2} = \frac{16 + 3}{500 + 100} = \frac{19}{600} = 0.03167$

Since $Q = 1 - P = 1 - 0.03167 = 0.96833$

$$\Rightarrow Q = 0.96833$$

$$e = \sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{(0.03167)(0.96833) \left(\frac{1}{500} + \frac{1}{100} \right)}$$

$$e = \sqrt{0.000368004} = 0.01918$$

$$Z = \frac{p_1 - p_2}{e} = \frac{0.032 - 0.03}{0.01918} = 0.104 < 1.96$$

As $|Z| < 1.96$, the hypothesis is accepted at 5% level of significance.

Hence we conclude that the machine has not improved after overhauling.

7) A machine produced 25 defective articles in a batch of 400. After overhauling it produced 15 defectives in a batch of 200. Test at 1% level of significance whether there is a reduction of defective articles after overhauling.

Soln: Null hypothesis: $H_0: P_1 = P_2$ i.e., the machine has not improved after overhauling.

Alternative hypothesis: $H_1: P_1 \neq P_2$.

Given $n_1 = 400$, $n_2 = 200$; $x_1 = 25$, $x_2 = 15$.

$$\therefore p_1 = \frac{25}{400} = 0.0625; \quad p_2 = \frac{15}{200} = 0.075$$

$$\text{WKT, } P = \frac{x_1 + x_2}{n_1 + n_2} = \frac{25 + 15}{400 + 200} = 0.0667$$

$$\text{Since } Q = 1 - P = 1 - 0.0667 = 0.9333$$

$$c = \sqrt{PQ \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} = \sqrt{(0.0667)(0.9333) \left(\frac{1}{400} + \frac{1}{200} \right)}$$

$$c = \sqrt{0.0004668} = 0.0216$$

$$Z = \frac{p_1 - p_2}{c} = \frac{0.0625 - 0.075}{0.0216} = -0.5787 < 2.58$$

As $|Z| < 2.58$, the hypothesis is accepted at 1% level of significance.

Hence we conclude that the machine has not improved after overhauling.

8) In an examination given to students at a large number of different schools the mean grade was 74.5 and standard deviation grade was 8. At one particular school where 200 students took the examination the mean grade was 75.9. Discuss the significance of this result at both 5% and 1% level of significance.

Soln: Given $n=200$, $\sigma=8$

$$\mu=74.5 \text{ and } \bar{x}=75.9$$

we calculate Z through Test statistic,

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{75.9 - 74.5}{8/\sqrt{200}} = \frac{1.4}{0.5657}$$

$$Z = \underline{\underline{2.47}}$$

(a) At 5% LOS:

As $|z| = 2.47 > 1.96$, the hypothesis is rejected at 5% level of significance.

(b) At 1% LOS:

As $|z| = 2.47 < 2.58$, the hypothesis is accepted at 1% level of significance.